FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Go Back

Back To Practice

Q1

Save

Completed

FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>

int main() {
    int n, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    int ref[n];

    printf("Enter the reference string: \n");
    for(int i = 0; i < n; i++)
        scanf("%d", &ref[i]);

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames];
    int pageFaults = 0, pageHits = 0;
    int index = 0;

    for(int i = 0; i < frames; i++)
        frame[i] = -1;

```

Test Cases 2

Run Sample Cases

Run All Cases

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Course Progress

Go Back

Back To Practice

Q1

Save

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Sample Input 1

Select Language : C (GCC 9.2.0)

```c

    for(int i = 0; i < n; i++) {
        int found = 0;

        for(int j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
                found = 1;
                pageHits++;
                break;
            }
        }

        if(!found) {
            frame[index] = ref[i];
            index = (index + 1) % frames;
            pageFaults++;
        }
    }

    printf("\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits: %d\n", pageHits);

    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Input :

```
7
1 3 0 3 5 6 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Q1

Save

Sample Test Case

Test Case - 1

Success

Input :

```
7
1 3 0 3 5 6 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Test Case - 2 Success

Input :

```
12
7 0 1 2 0 3 0 4 2 3 0 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 10
Total Page Hits: 2
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 10
Total Page Hits: 2
```